

Algebra II with Trigonometry

Chapter 6.2

Olympiad Problems (Gemini)

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Problems

1. Suppose t is a real number such that $\sec t + \tan t = 2024$. Evaluate the exact value of $\sec(-t) + \tan(-t)$.

2. Let $f(x) = \frac{x^3 \sec x}{1 + \cos^2 x} - 5 \csc(x) \sin^2(x) + 10$. Evaluate the exact sum of $f\left(-\frac{\pi}{5}\right) + f\left(\frac{\pi}{5}\right)$.

3. Evaluate the exact value of the following nested composition:

$$\tan\left(\frac{\pi}{4} \sec\left(\frac{\pi}{3} \cos\left(\pi \sin\left(\frac{7\pi}{6}\right)\right)\right)\right)$$

4. The real number t satisfies the inequality $\tan t + \cot t < 0$. If it is also known that $\sec t > \csc t$, determine the quadrant in which the terminal point of t lies.

5. For the real number $t = \frac{2024\pi}{13}$, determine the exact value of the expression:

$$\sec^4 t - \tan^4 t - 2 \tan^2 t$$

6. A real number t satisfies $\sin t + \cos t = \frac{1}{3}$. Determine the exact value of $\tan t + \cot t$.